Project B: University Services Management Platform

Case Scenario and Scope Specification

Module	Software Architecture and Process Models
Student Level	Level 2
Assignment Duration	7 weeks
Document	03 of 07
Assigned Groups	Groups 5, 6, 7, and 8
Product Type	Integrated web-based functional prototype

1. Case Scenario

A university provides many services through separate offices, paper forms, email, spreadsheets, and informal messaging channels. Students and staff may need to reserve lecture halls, laboratories, meeting rooms, or equipment; report facility or IT problems; register for events; receive announcements; and track requests. Service staff and administrators need a consistent way to prioritize work, manage availability, communicate updates, and review service performance.

The project must deliver a University Services Management Platform that brings these activities together without attempting to replace the university student information system, learning management system, finance system, or human resource system. The platform should use synthetic user data and demonstrate secure, role-based, integrated services.

2. Project Goal

> **Goal statement**
> Design, implement, test, integrate, and deploy a secure functional prototype through which students, academic staff, administrative staff, and service personnel can access and manage selected university services.

3. Primary Users and Roles

Role	Typical Responsibilities
System Administrator	Manage platform roles, access, reference data, and system-level visibility.
Student	View services, reserve eligible resources, submit requests, register for events, receive announcements, and provide feedback.
Academic Staff	Reserve teaching or meeting resources, submit requests, organize events, and receive targeted communications.
Administrative Staff	Manage service information, approve reservations, publish announcements, coordinate events, and review summaries.
Service Desk Officer	Review incoming service requests, categorize and prioritize them, communicate updates, and coordinate work.
Technician / Maintenance Staff	View assigned work orders, update progress, record actions, and close completed work.
Resource Manager	Maintain facilities/equipment, rules, availability, reservations, and approvals.
Event Organizer	Create events, control capacity and eligibility, manage registrations, and view attendance-related summaries.

4. Team Domain Allocation

Group	Domain Ownership	Owned Microservices	Frontend Feature Area
Group 5	Identity, access, university directory, and role relationships	identity-access-service; university-directory-service	Login, profile, role-aware navigation, directory and department/service-unit administration.
Group 6	Facilities, equipment, availability, and reservations	facility-resource-service; reservation-service	Resource search, calendars, reservations, approvals, cancellations, and utilization views.

Group	Domain Ownership	Owned Microservices	Frontend Feature Area
Group 7	Service requests, incidents, maintenance, and work orders	service-request-service; work-order-service	Request submission, triage, assignment, status tracking, technician workspace, and service summaries.
Group 8	Events, announcements, feedback, and engagement	event-service; communication-feedback-service	Events, registrations, announcements, notifications, feedback, and engagement dashboards.

The API Gateway, shared frontend shell, Docker Compose integration, and common release documentation are jointly governed by all four teams through the Project Integration Council.

5. Group 5 Scope - Identity and University Directory

- Authenticate users, issue JWTs, and enforce role-based authorization for students, staff, administrators, and service personnel.
- Maintain synthetic user profiles with university identifier, role, department or service unit, status, and contact information.
- Maintain departments, faculties, service units, and role relationships needed by other services.
- Allow authorized administrators to activate, deactivate, and update user access and role assignments.
- Expose APIs that allow other services to validate users, roles, departments, and service responsibilities.
- Support role-aware navigation and profile views in the shared frontend.

Key business rule	The platform must not assume that every staff member has the same permissions. Authorization must consider both system role and relevant department or service responsibility.

6. Group 6 Scope - Facilities and Reservations

- Manage facilities and resources such as lecture halls, laboratories, meeting rooms, sports spaces, and equipment.
- Maintain location, capacity, resource type, operating hours, availability status, eligibility rules, and approval requirements.
- Allow eligible users to search availability and submit reservation requests for a date and time period.
- Prevent conflicting approved reservations and enforce capacity, operating-hour, and eligibility constraints.
- Support approval, rejection, cancellation, check-in or usage status where appropriate, and reservation history.
- Provide calendar and utilization summaries for resource managers.

Key business rule	A reservation cannot be approved if the resource is unavailable, the time conflicts with an approved reservation, or the requester is not eligible under the configured rule.

7. Group 7 Scope - Service Requests and Work Orders

- Allow students and staff to submit facility, equipment, IT, or general service requests with category, location, priority suggestion, description, and attachment reference.
- Allow service desk staff to triage, classify, prioritize, assign, escalate, or reject requests with reasons.
- Create and manage work orders for technicians or service teams, including assignment, schedule, status, action notes, and resolution.
- Allow requesters to view progress and provide confirmation or feedback after completion.
- Track service-level timestamps such as reported, acknowledged, assigned, started, resolved, and closed.
- Provide request and work-order summaries by category, status, priority, location, and responsible service unit.

Key business rule	Only an authorized service desk or assigned technician may perform protected status changes. Closure requires a recorded resolution, and rejected or cancelled requests require a reason.

8. Group 8 Scope - Events, Communications, and Feedback

- Create and publish university events with organizer, venue or online mode, schedule, capacity, eligibility, registration period, and status.
- Allow eligible users to register, cancel, and view registration status while preventing capacity over-allocation.
- Publish announcements targeted by role, department, faculty, service unit, or all users.
- Provide in-application notifications for relevant reservation, service-request, event, and announcement updates.
- Collect feedback or ratings for selected events or services using configurable forms or questions.
- Provide event participation, announcement reach, and feedback summaries appropriate to a prototype.

Key business rule	Registration must close when capacity is reached, the registration period ends, or the event is cancelled. Users who are not eligible must receive a clear explanation.

9. Required Cross-Team Workflows

Workflow	Minimum Cross-Service Interaction
Authenticated service access	Identity and directory data determine which services, resources, and administrative actions a user may access.
Resource reservation	User eligibility is validated; facility availability is checked; reservation is created; user receives status notification.
Maintenance request for a facility	User and facility are validated; request is triaged; work order is assigned; requester sees progress and resolution.
Event with physical venue	Organizer and venue are validated; event capacity and schedule are set; registration and notifications are managed.
Targeted announcement	Authorized staff select audience using role or department data; announcement is published and visible to the intended users.
Service feedback	A completed request or event becomes eligible for feedback; results are summarized for authorized administrators.

10. Core Entities

Domain	Indicative Entities
Identity and directory	User, Role, Permission, UserRole, Department, Faculty, ServiceUnit, UserAffiliation.
Facilities and reservations	Facility, Resource, ResourceType, AvailabilityRule, Reservation, ReservationApproval, TimeSlot.
Service operations	ServiceRequest, RequestCategory, Priority, WorkOrder, Assignment, ActionLog, Resolution.
Events and engagement	Event, EventEligibility, Registration, Announcement, AudienceRule, Notification, FeedbackForm, FeedbackResponse.

Teams must refine these entities and may combine or separate them when justified. The final design must preserve service data ownership and avoid a shared monolithic database.

11. System-Wide Non-Functional Requirements

- Security: JWT-based access, role and responsibility checks, secure password storage, validated requests, and safe error messages.
- Privacy: synthetic data only; do not use actual student or staff personal records.
- Usability: responsive, accessible, consistent navigation with clear status and validation feedback.
- Reliability: dependent-service failures must be handled without corrupting local records or misleading users.
- Performance: resource search, request tracking, and event list operations should remain responsive under normal classroom-scale load; observations must be reported.
- Maintainability: clear service ownership, documented API contracts, repository standards, automated tests, and configuration examples.
- Auditability: administrative actions, approvals, protected status changes, and important timestamps must be traceable.

12. Minimum Reports and Dashboards

- Resource availability and utilization summary.
- Reservations by status, resource type, and period.
- Service requests and work orders by category, priority, status, and service unit.
- Upcoming events, registrations, and capacity summary.
- Announcements and recent notifications.
- Role-appropriate personal dashboard for student, staff member, resource manager, service officer, or technician.

13. Out of Scope

- Replacement of the university student information system, learning management system, HR system, finance system, or library system.
- Connection to real university identity, student, staff, timetable, or payroll databases.
- Real payment processing, biometric access, physical smart-card control, or IoT equipment integration.
- Native mobile application.
- Real SMS or email delivery; in-application or simulated notifications are sufficient.
- Commercial deployment, live operational use, warranty, or production support obligations.

14. Acceptance Conditions

[ ] All four team domains are accessible through one shared frontend and API Gateway.

[ ] At least two end-to-end workflows cross service or team boundaries and are demonstrated successfully.

[ ] Authentication, authorization, validation, error handling, data persistence, and logging are demonstrated.

[ ] Each owned service has Swagger/OpenAPI documentation, database scripts, automated tests, and a reproducible local setup.

[ ] The integrated release can be started locally using documented Docker Compose steps and is also deployed to an approved environment.

[ ] Jira, GitHub, testing, release, and project closure evidence is complete.